ChemPubSoc Europe
DOI: 10.1002/slct.201801565
Chemistry SELECT Full Papers

Organic & Supramolecular Chemistry

Autoxidation of Siphonochilone in Processed Rhizomes and Stored Powders of *Siphonochilus aethiopicus* (Schweinf.) B.L. Burtt

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A furano sesquiterpenoid, known as siphonochilone (**1**), is commonly reported as the major compound in the roots and the rhizomes of *Siphonochilus aethiopicus* together with related oxidized sesquiterpenoids (**2**-**5**). As part of investigating the UPLC QTOF MS ES+ chemical profiles of n-hexane/dichloromethane (1:1) extracts from crushed fresh rhizomes, dried and powdered rhizomes and the dried powders held under storage at room temperature, we have recently realized that decomposition of **1** occurs during the processing of the rhizomes and the storage of the powders. This decomposition which involves autoxidation of the furan moiety leads to formation of the artefact sesquiterpenoid lactone (**3**) and potentially (**4**), which numerous papers misreport as naturally present in the plant. MP2/6-31+G(d,p), B3LYP/6-31+G(d,p) and B3LYP/6-311+G(2d,p) calculations were performed on siphonochilone (**1**) and the sesquiterpenoid lactone artifacts (**3**) and (**4**) and results rationalized the observed autoxidation.

Introduction

Siphonochilus aethiopicus (Schweinf.) B.L. Burtt, also known as African ginger, is a deciduous rhizomatous species in the Zingiberaceae family and indigenous to savanna regions of tropical and southern Africa.[1–3] The rhizomes of the plant are highly prized in the South African traditional medicine for its health benefits where it is widely employed as a local remedy against a number of ailments such as asthma, colds, coughs, hysteria and influenza.[4–8] In 2002, two furano sesquiterpenoids, viz. siphonochilone (**1**) and its hydroxylated derivative (**2**), and their relative configurations were reported by Holzapfel et al.[3] after steam distillation of fleshy crushed roots while three aditional sesquiterpenoid lactones (**3**-**5**) were purified and reported in 2009 from an ethylacetate extract of dried and powdered rhizomes by Lategan et al.[5] The relative structures are shown in Figure 1.

In our studies we aimed at investigating the chemical profiles using ultra performance liquid chromatography coupled to mass spectrometry (UPLC QTOF MS ES+) of n-hexane/dichloromethane (1:1) extracts prepared from crushed fresh rhizomes, processed and powdered rhizomes and powders stored under room temperature conditions for nine months. We have observed that levels of siphonochilone (1) reduce through oxidation of its furan moiety upon storage, leading to formation of more stable sesquiterpenoid lactones (3) and (4).

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Supporting information for this article is available on the WWW under https://doi.org/10.1002/slct.201801565

Results and Discussion

Autoxidation of siphonochilone after plant processing and storage of the powder

Dried n-hexane/dichloromethane (1:1) extracts from crushed fresh rhizomes, freshly processed and powdered rhizomes and from powders stored for nine months at room temperature were subjected to UPLC QTOF MS analyses using positive electrospray ionization (ES+). Results of the extract from the fresh rhizomes are expressed as chromatographic chemical profiles of the secondary metabolites and four characteristic peaks, labeled A–D, were annotated (Figure 2a). The most intense peak D was identified as siphonochilone **1** after purification – as a colorless crystalline compound (Figure 2b), and molecular characterization through MS, NMR, IR, X-ray diffraction experiments (XRD), and B3LYP/6-311+G(2d,p)[9] structure optimization.

QTOF MS ES+ data from the UPLC elution of the isolated siphonochilone **1** displayed a $[M+H]^+$ molecular ion with m/z 231.1387 (Figure 2b) which corresponded to the molecular formula $C_{15}H_{18}O_2$. Intensive NMR investigations established the relative configuration of **1**, which latter corroborated the XRD structure (Figure 3a) as well the B3LYP/6-311+G(2d,p) optimized geometry (Figure 3b).

Comparison between XRD structural data and B3LYP/6-311+G(2d,p) geometrical parameters validated the identity of

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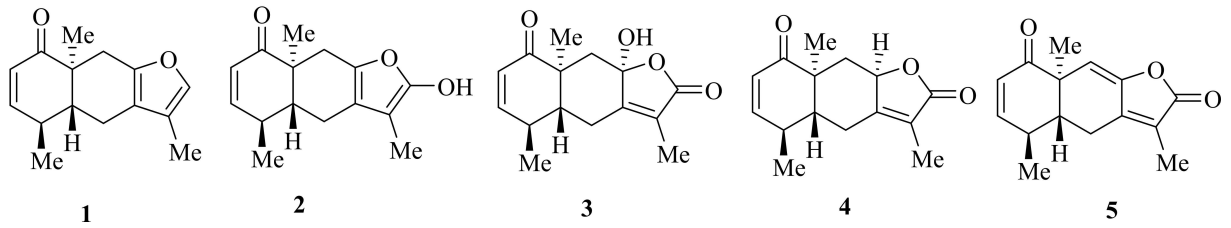

Figure 1. Siphochilone (**1**), its hydroxylated derivative (**2**) and related sesquiterpenoid lactones (**3-5**).

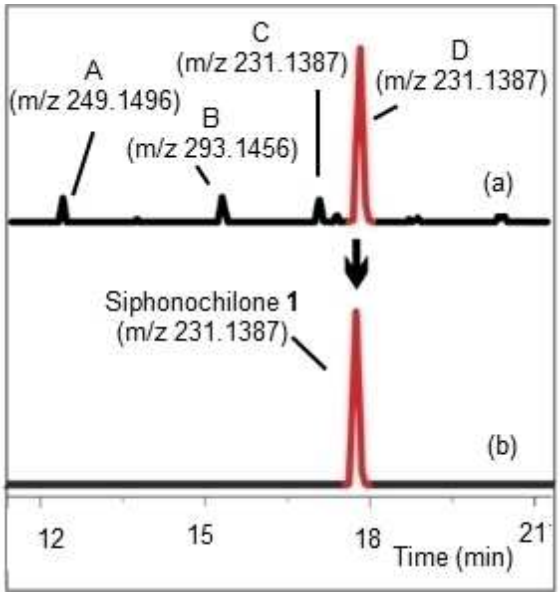

Figure 2. (a) UPLC QTOF MS ES + chemical profile of the extract prepared from fresh rhizomes (b) UPLC QTOF MS ES + chromatogram of the isolated siphonochilone **1.**

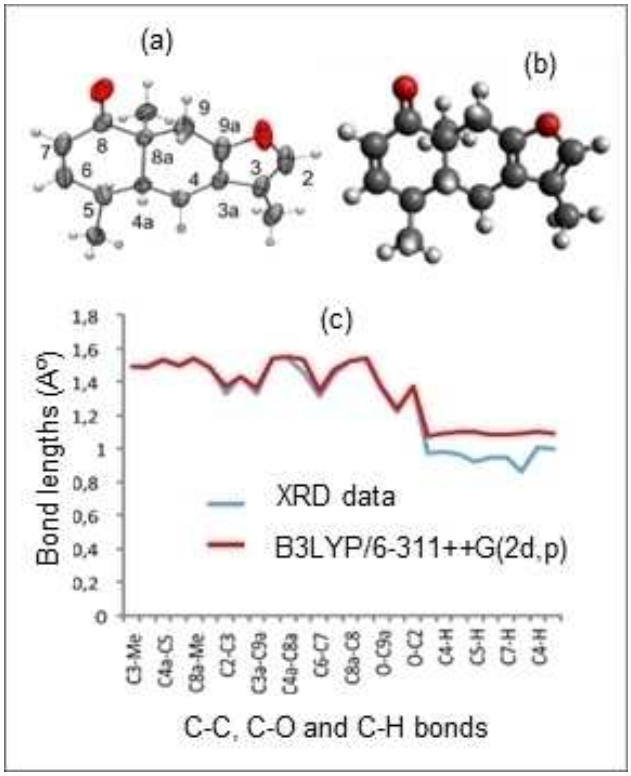

Figure 3. (a) X-ray crystallography structure (b) B3LYP/6-311 + +G(2d,p) optimized geometry of siphonochilone **1 (c)** Comparison between bond distances of the B3LYP/6-311 + +G(2d,p) optimized geometry (red curve) and X-ray geometrical data (blue curve) of siphonochilone **1.**

siphonochilone **1**. The computed bond distances matched perfectly with experimental data (Figure 3c). Small deviations were however observed for C–H bonds. This shortening of bonds in the experimental XRD geometry commonly arises from the arrangement of molecules within the crystalline solid.[10] This may also be a result of incorrectly defined proton positions in the XRD experiment – an often encountered effect.

Table 1 and Table 2 compare the NMR experimental data with the results derived from B3LYP/6-311 + G(2d,p) quantum chemical calculations in dichloromethane.

The similarity between the experimental and theoretical ^{13}C and ^{1}H NMR chemical shifts relative to TMS as well as the ^{1}H spin–spin coupling (Table 1 and 2) constants undoubtedly proved the isolated compound to be siphonochilone (**1**).

Further validation of the molecular structure of siphonochilone arose from the matching of theoretical and experimental vibrational frequencies, which were found to be in good agreement despite minor shifts (Table 3). These shifts in frequencies give a glimpse on the strength of the intermolecular interactions in the crystalline solid of siphonochilone **(1)** and show the effect of the media in the solid state IR, referring to the fact that the theoretical IR spectrum was achieved in gas phase. It is known that C–C bonds are more rigid and constitute the backbone of the molecular system on which other pendant functions such as C–H are linked. The pendant groups are more flexible and can be involved in intermolecular interaction with neighbouring molecules in condensed phase, hence causing vibrational frequencies shifts.[10]

The three other peaks marked as A, B and C (Figure 2a) could not unambiguously be related to any of the sesquiterpenoids reported from African ginger and shown in Figure 1.

Investigation of the n-hexane/dichloromethane (1:1) extract prepared from the powder obtained right after processing the sliced, dried and grinded rhizomes displayed a chemical profile (Figure 4b) similar to that from the fresh plants (Figure 4a), with siphonochilone being the major peak. However, an additional minor peak could be detected in the profile from the processed

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Table 1. NMR Spectroscopic Data (400 MHz, CD_2Cl_2) for siphonochilone (**1**). Labels refer to the X-ray structure of **1** in Figure 3a. B3LYP/6-311+G(2d,p) chemical shifts relative to TMS in dichloromethane are shown in parenthesis.

Position	δ_C, type	δ_H	HMBC
2	137.8 (142.6), CH	7.1, *m* (7.3)	
3	119.7 (128.2),C		
3a	115.3 (121.4),C		
4	22.8 (25.6), CH_2	2.7, *ddd* (2.8), 2.1, *dddd* (2.2)	3a, 4a, 9a, 3-Me
4a	45.3 (49.7), CH	1.9, *ddd* (1.8)	4, 5, 8a, 8a-Me
5	34.6 (39.4), CH	2.4, *m* (2.6)	
6	154.8 (166.3), CH	6.7, *dd* (7.2)	4a, 5, 8, 5-Me
7	126.7 (131.1), CH	5.9, *dd* (6.1)	5
8	204.2 (214.5), C		
8a	45.3 (50.9), C		
9	32.3 (34.7), CH_2	2.7, *dd* (2.7), 2.6, *br d* (2.6)	3a, 4a, 9a, 8a-Me, 9a, 8a-Me
9a	149.6 (157.7),C		
3-Me	8.2 (8.8), CH_3	1.9, *d* (1.9)	2, 3, 3a
5-Me	18.9 (18.9), CH_3	1.2, *d* (1.3)	4a, 5, 6
8a-Me	16.8 (17.3), CH_3	1.0, *s* (1.1)	4a, 8, 9

Table 2. Experimental 1H-1H spin spin coupling constants (400 MHz, CD_2Cl_2) for siphonochilone (**1**) and that computed at B3LYP/6-311+G(2d, p) in dichloromethane.

Position	Type	Exp.(Hz)	B3LYP/6-311+G(3d, 2p)
4	CH_2	(15.9, 5.4, 1.8) (15.9, 11.0, 3.1, 1.7)	(15.5, 5.4, 2.0) (15.5,9.4,3.5,1.6)
4a	CH	(10.9, 9.9, 5.4)	(9.4, 8.0, 5.4)
6	CH	(10.1, 2.2)	(9.4, 2.4)
7	CH	(10.1, 2.8)	(9.4, 3.4)
9	CH_2	(16.8, 1.7) (16.8)	(16.7, 1.9) (16.7, 1.6)
3-Me	CH_3	(1.3)	(2.2)
5-Me	CH_3	(7.2)	(11.7)
8a-Me	CH_3		

Table 3. Assignment of different stretching vibrational modes of **1** computed at B3LYP/6-311+G(2d,p) *in vacuo* and that carried out experimentally in the solid state.

Experimental frequency	Theoretical frequency	shifts	Assignments	Intensities
3131-3019	3261-3136	44-130	ν C–H	av,w
2968	3109-3087	119-141	ν CH_3	s
2932-2849	3071-3051	139-202	ν C H_2	s
1759	1698	61	ν C=O	s
1673-1646	1677-1589	4-57	ν C=C	av,w

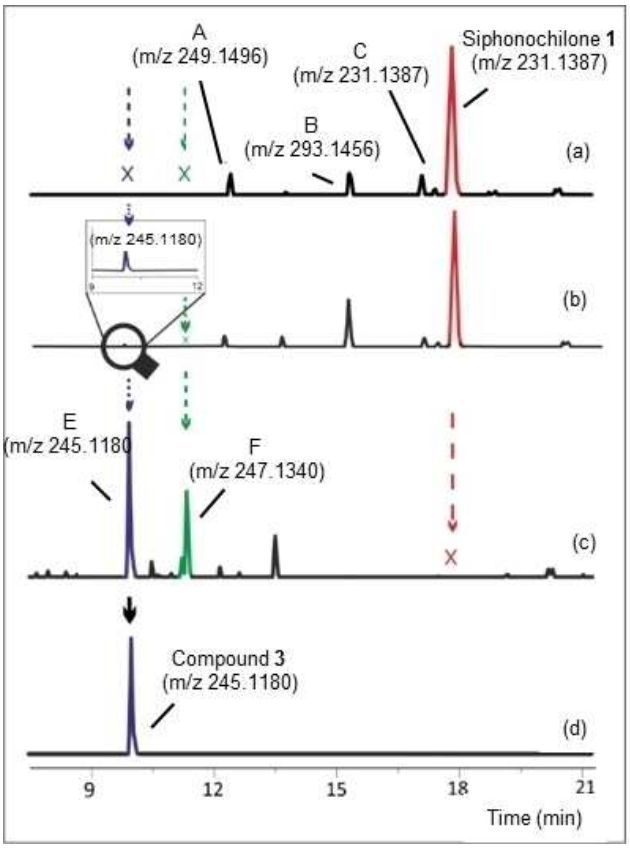

Figure 4. UPLC MS ES+ chemical profiles of extracts from (a) fresh rhizomes, (b) processed plants, (c) stored powder (d) chromatograms of the pure compound **3**.

plants at 10.03 min (m/z 245.1180), see the purple peak in the zoomed region in Figure 4b. This peak was observed to intensify in the chromatogram of the extract prepared from the stored powder (see the purple peak in Figure 4c). The latter chromatogram revealed three characteristic peaks while lacking a peak corresponding to siphonochilone. Of these three peaks, two labelled E and F were newly formed, as they were completely absent in the chromatogram of the fresh rhizomes (Figure 4a). The compound which corresponds to peak E was unambiguously identified after isolation, crystallization and structure elucidation as the sesquiterpenoid lactone **3**.

UPLC QTOF MS ES+ analysis of the isolated compound **3** showed a single peak (Figure 4d) with a $[M+H]^+$ corresponding to *m/z* 263.1282 and an abundant base peak of *m/z* 245.1180 $[M+H\text{-}H_2O]^+$. The $[M+H]^+$ molecular ion (*m/z* 263.1282) was assigned the formula $C_{15}H_{18}O_4$.

The relative molecular structure was derived from NMR analyses, XRD experiments and B3LYP/6-311+G(3d,2p) theoretical calculations. Table 4 and Table 5 provide B3LYP/6-311+G(2d,p) data of ^{13}C and ^{1}H NMR chemical shifts relative to TMS as well as the ^{1}H spin–spin coupling in chloroform.

Comparison between structural data from XRD and the molecular parameters issued from B3LYP/6-311+G(2d,p) calculations for the sesquiterpenoid lactone **3** showed a good match

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Table 4. NMR Spectroscopic Data (400 MHz, CD_3Cl) for compound **3**. Labels refer to the X-ray structure of **3** in Figure 5a. B3LYP/6-311+G(2d,p) chemical shifts relative to TMS in chloroform are shown in parenthesis.

Position	δC, type	δH	HMBC
2	172.1 (178.4), C		
3	122.4 (130.0), C		
3a	158.5 (169.7),C		
4	24.1 (27.0), CH_2	2.8, dd (2.9), 2.4, m (2.4)	3, 4a, 8a, 9a, 3-Me
4a	49.9 (55.6), CH	1.6, *ddd* (1.5)	8a-Me
5	33.8 (39.2), CH	2.5, *dqt* (2.6)	
6	153.6 (164.2), CH	6.6, *dd* (7.1)	4a, 5, 8, 5-Me
7	126.1 (131.3), CH	5.9, *dd* (6.0)	5
8	202.9 (212.6), C		
8a	44.9 (51.2), C		
9	43.7 (46.2), CH_2	2.6, *d* (2.7), 1.7, *d* (1.5)	3a, 4a, 9a, 8a, 9a, 9a, 8a, 8a-Me
9a	103.5 (110.0), C		
9a-OH		4.1, *s* (1.4)	
3-Me	8.2 (9.1), CH_3	1.8, *d* (1.8)	2, 3, 3a
5-Me	18.1 (18.5), CH_3	1.2, *d* (1.3)	4a, 5, 6
8a-Me	16.7 (17.6), CH_3	1.3, *s* (1.4)	4a, 8, 8a

Table 5. Experimental 1H-1H spin spin coupling constants (400 MHz, CD_3Cl) for compound **3** and that computed at B3LYP/6-311+G(2d, p) in chloroform.

Position	Type	Exp.(Hz)	B3LYP/6-311+G(3d, 2p)
4	CH_2	(13.5, 3.6)	(12.9, 3.4), (12.9, 11.0, 2.7)
4a	CH	(13.3, 10.0, 3.6)	(11.0, 3.4, 3.0)
6	CH	(10.1, 2.0)	(9.4, 3.4)
7	CH	(10.1, 2.7)	(9.4, 3.4)
9	CH_2	(14.5), (14.5)	(14.1), (14.1)
3-Me	CH_3	(1.4)	(2.7)
5-Me	CH_3	(7.1)	(11.6)
8a-Me	CH_3		

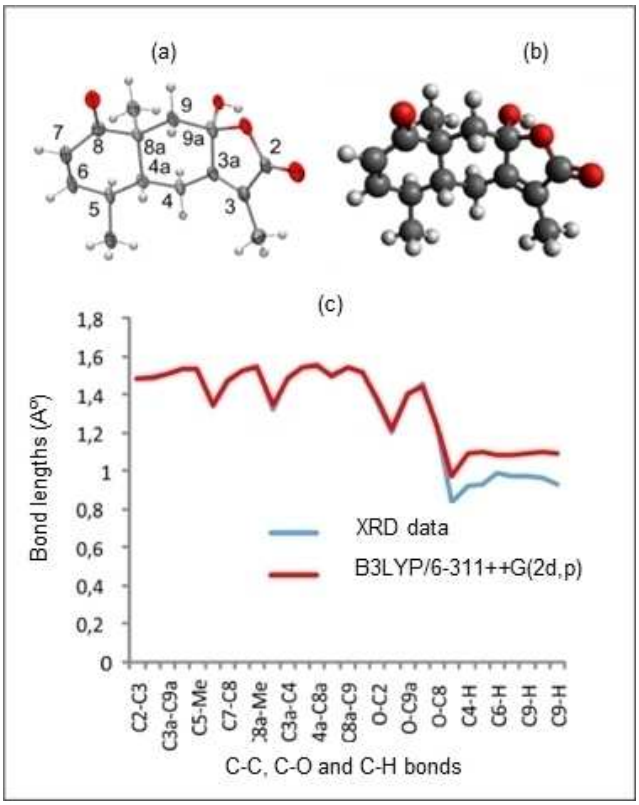

Figure 5. (a) XRD crystallographic structure of **3** (b) B3LYP/6-311++G(2d,p) optimized geometry (c). Comparison between the bond distances of the B3LYP/6-311++G(2d,p) optimized geometry (red) and X-ray geometry (blue) of sesquiterpenoid lactone **3.**

Table 6. Assignment of different stretching vibrational modes of **3** computed at B3LYP/6-311+G(2d,p) *in vacuo* and that carried out experimentally in the solid state.

Experimental frequencies	Theoretical frequencies	shifts	Assignments	Intensities
3425	3794	369	ν O-H	s
3131-3019	3197-3150	44-117	ν C-H	av,w
2968	3152-3125	93	ν CH_3	s
2932-2849	3078-3017	119-168	ν CH_2	s
1759-1673	1836-1747	77-74	ν C=O	vs
1646	1743-1681	4-57	ν C=C	av,w

between experimental and predicted bond lengths with small deviations for C–H and O–H distances (Figure 5c). The observed deviations could be explained as a result of O–H...O intramolecular hydrogen bond (IHB) formation in the crystal structure, between the hydroxyl group and its neighboring oxygen atom of the five-membered ring. Both B3LYP/6-31+G(d,p) and MP2/6-31+G(d,p)[9b,11] showed this intramolecular O–H...O bond to provide a small stabilization energy of 0.13 kcal/mol and 0.16 kcal/mol, respectively, compared to the geometric structure optimized without consideration of the O–H...O IHB.

Good correlation was also observed between the solid state experimental IR data and the B3LYP/6-311+G(2d,p) harmonic frequencies (Table 6). The shifts in frequencies could be depicted as a result of intra and intermolecular interactions arising within the solid crystal of the sesquiterpenoid lactone **3**. C–H, O–H, and C=O groups can in fact interact with neighbouring molecules through intermolecular hydrogen bonds. The largest redshifts observed for O–H and C–H was in good agreement with the bond lengths difference displayed in Figure 5c.

The peak labeled F (Figure 4c) showed a molecular ion $[M+H]^+$ with m/z 247.1338 which corresponded the molecular formula $C_{15}H_{18}O_3$. This formula matched those of the sesquiterpenoids (**2**) and (**4**). After isolation and characterization of its

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molecular structure, the peak F was conclusively identified as compound **4**. This was done through comparison of the acquired NMR data with the ^{1}H and ^{13}C chemical shifts reported in the literature (see supporting information) as well as the data predicted at the B3LYP/6-311+G(2d,p) theoretical level. These results reliably indicated that the two sesquiterpenoid lactones **3** and **4** originated from the autoxidation of siphonochilone (**1**) and are formed during storage of the dry powder.

Autoxidation of the purified siphonochilone

To rationalize the conversion of siphonochilone **1** into the more oxidized sesquiterpenoid lactones **3** and **4** during storage, UPLC QTOF MS ES+ investigation was conducted on the isolated and stored siphonochilone. The crystals was stored in transparent glass vials at room temperature and visually monitored on a regular basis for 5 months. The initially colorless crystals were gradually observed to turn brownish on standing. UPLC QTOF MS ES+ investigation of the brownish crystals showed two additional peaks, in addition to that of siphonochilone **1** (Figure 6b). Comparison between the generated chromatographic data with the chemical profile of the extract prepared from the powder held under storage (Figure 6c) showed that the two new peaks – labeled E' and F', to be the same as those observed for the stored powder (Figure 6b versus Figure 6c), giving that retention times as well as MS data were in perfect agreement. From this matching, the identity of E' was thus reliably confirmed to be the hydroxy lactone derivative **3**, while that of F' was established to be **4** (supplementary information).

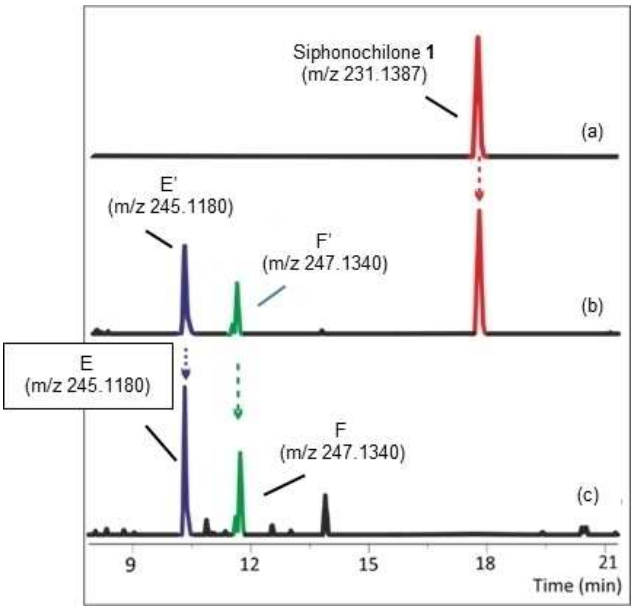

Figure 6. UPLC MS ES+ chromatogram of siphonochilone (a), siphonochilone after autoxidation (b) and stored powder (c).

The above results clearly indicated that autoxidation of siphonochilone occurs in the stored powder as well as in the isolated crystals held in the glass transparent vial, leading to formation of more oxidized artefacts, such as the lactones **3** and **4**.

Rationalization of the autoxidation of siphonochilone **1** into the more oxidized artefacts was achieved through an intensive study of HOMO and LUMO orbitals of compounds **1**–**4**. Computations were carried out at MP2/6-31+G(d,p) level and results are portrayed in Figure 7 for compounds **1**–**4**. Results estimated the HOMO-LUMO gap energy of **2** to be about 9.35 eV, that of **3** as 11.60 eV, while that of **4** was found to be approximately 11.29 eV. Comparison between the HOMO-LUMO gap energies of the compounds **3** and **4** with that of siphonochilone **1** which was established as 9.66 eV, revealed the more oxidized compounds **3** and **4** as more kinetically stable entities, given their greater HOMO-LUMO gaps while

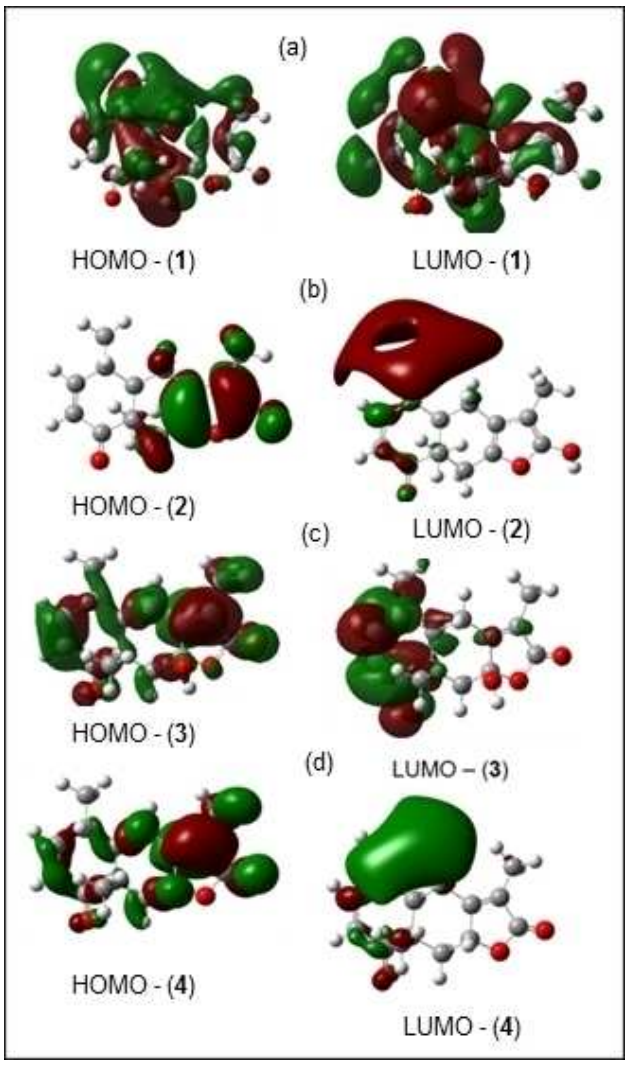

Figure 7. Frontier molecular orbitals of siphonochilone **1** and its structurally related sesquiterpenoids **2**–**4**.

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that for compound **2** is close to that of **1**. These results might justify the observed autoxidation of siphonochilone **1** to form chemically more stable artefacts such as **3** and **4,** fact being that the difference between the HOMO-LUMO gap potential energies is a driving force of different types of chemical reactions.

Conclusions

A furano sesquiterpenoid, known as siphonochilone **1**, is commonly reported as the major compound in the roots and the rhizomes of African ginger together with closely related sesquiterpenoid derivatives (**2**-**5**). We demonstrated in this study that autoxidation of siphonochlone **1** occurs during the processing of the rhizomes and the storage of the powders, leading to formation of artifact lactone derivatives **3** and **4** which numerous papers might be misreporting as naturally present in the African ginger rhizomes. MP2/6-31+G(d,p) theoretical investigation of the frontier molecular orbitals as well as the HOMO-LUMO potential energy gaps proved the artefact **3** and **4** to be more chemically stable than **1**, hence justifying autoxidation of siphonochilone **1** into more stable compounds.

The experimental geometries and spectroscopic measurements (IR, NMR) of the least and most oxidized compounds (**1** and **3**) were found in good agreement with B3LYP and MP2 results.

Supporting information summary

Experimental (extraction, purification and isolation procedures), NMR data and computational details are given in the supporting information.

Acknowledgements

The Department of Enterprise Creation (ECD) at the South African Council for Scientific and Industrial Research (CSIR) is acknowledged for the supply of African ginger rhizomes used for the present study. JT Muya thanks the Korea Research Fellowship program and grants from the Basic Science Research Program funded by the Ministry of Science, ICT and Future planning through the National Research Foundation of Korea (NRF2015H1D3 A1062502) for support. FZJ Katele acknowledges BEBUC (Bourse d'Excellence Bringmann aux Universités Congolaises) and fUNIKIN (foerdeverein Uni Kinshasa e.V.) for the MSc scholarship, as well as and the Holger Pöhlmann Stiftung for support.

Conflict of Interest

The authors declare no conflict of interest.

Keywords: African ginger · autoxidation · siphonochilone · Siphonochilus aethiopicus

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Submitted: May 24, 2018
Accepted: July 20, 2018

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